International Review of Financial Analysis 93 (2024) 103164

Contents lists available at ScienceDirect

International Review of Financial Analysis

journal homepage: www.elsevier.com/locate/irfa

Data breach disclosures and stock price crash risk: Evidence from data breach notification laws

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ARTICLE INFO

JEL classification:
G10
G30
G38

Keywords:
Data security
Data breaches
Disclosure
Stock price crash risk
Bad news hoarding

ABSTRACT

Exploiting the staggered adoption of data breach notification (DBN) laws, which obligate firms to disclose data breaches when they occur, as an exogenous shock to data breach disclosures, we find that the adoption of these laws leads to higher future stock price crash risk. The positive relation between DBN laws and crash risk is more pronounced for firms with weaker corporate governance, higher financial constraints, and higher information asymmetry. Our findings suggest that investors' concerns about the consequences of data breaches and the vulnerability of breached firms' data security heighten stock price crash risk.

1. Introduction

Data breaches have surged over the last few decades, with hundreds of millions of consumer records compromised and thousands of breaches reported each year.[1] A stream of literature has examined the impacts of data breaches on various stakeholders, from customers to employees to suppliers. Identity theft has serious effects on consumers, leaving about "14% of the victims with out-of-pocket losses and 36% with emotional distress" (Harrell & Langton, 2013). Data breaches are costly to the affected firms, damaging their reputation, decreasing sales, and even leading to a loss of business (Gatzlaff & McCullough, 2010; Lending, Minnick, & Schorno, 2018). These incidents also deteriorate firms' credit ratings and make their cash flows more volatile (Campbell, Gordon, Loeb, & Zhou, 2003; Mikhed & Vogan, 2018; Kamiya, Kang, Kim, Milidonis, & Stulz, 2021; Huang & Wang, 2021). A recent study by IBM Security reports that the average cost per data breach to U.S. organizations reached a record high of $9.44 million in 2022.[2]

Corporate managers are likely to be sensitive to the adverse effects of data breach disclosures on their reputation, job safety, and compensation, therefore, they may hoard bad news about data breaches or time their disclosures strategically to mitigate the consequences (Foerderer & Schuetz, 2022; Xu, Guo, Haislip, & Pinsker, 2019). However, when accumulated bad news reaches a critical threshold level, managers cannot hide it any longer and will have to disclose such news, potentially leading to stock price crash, a large negative outlier in the stock return distribution. In this study, we ask whether and how data breach disclosures affect stock price crash risk (hereafter crash risk).

Because managers may choose to withhold bad news related to data breaches and time their disclosures, possible self-selection bias presents a challenge to an empirical analysis of the relation between data breach disclosures and crash risk. It is worth noting that in response to the growing frequency and severity of data breaches, U.S. states have adopted data breach notification (DBN) laws, which mandate data breach reporting, to simultaneously mitigate the risks posed by data breaches to customers while encouraging businesses to improve cybersecurity practices. Romanosky, Telang, & Acquisti (2011) report that the number of data breach disclosures increases significantly following the passage of DBN laws. The disclosure obligation induces firms to increase investment in cybersecurity to reduce the likelihood of future data breaches and rebuild customer confidence (Romanosky et al., 2011;

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[1] ITRC 's, 2021 Annual Data Breach Report.
[2] Available at https://www.ibm.com/reports/data-breach. Last accessed on December 7, 2022.

https://doi.org/10.1016/j.irfa.2024.103164
Received 22 February 2023; Received in revised form 17 December 2023; Accepted 6 March 2024
Available online 8 March 2024
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Sullivan & Maniff, 2016). In this study, we exploit the staggered adoption of DBN laws by various U.S. states between 2002 and 2018 as a quasi-natural experiment to identify the relation between data breach disclosures and crash risk.

In 2002, California was the first state to enact DBN law that requires breached firms to notify authorities and affected customers when "unencrypted personal information was, or is reasonably believed to have been, acquired by an unauthorized person" (Cal. Civ. Code § 1798.82). As of March 29, 2018, all 50 U.S. states and the District of Columbia, had enacted DBN laws requiring private businesses and government entities to notify authorities and affected individuals of security breaches. The staggered adoption of DBN laws allows us to employ the difference-in-differences (DiD) approach, which compares the change in the likelihood of stock price crash of firms headquartered in states that adopt DBN laws (treatment firms) and firms headquartered in states that do not adopt DBN laws (control firms).

Although specific disclosure requirements vary across states, the primary objective of state-level DBN laws is to mandate data breach disclosures, thereby promoting public awareness and personal internet security (Romanosky et al., 2011). These laws diminish managers' ability to hide breach news from the public and improve corporate transparency, which potentially reduces future crash risk. However, following the adoption of DBN laws, managers have to disclose not only data breaches as they happen but also those that accumulated from before the passage of DBN laws. These disclosures suggest not only past data insecurity at the breached firms but also their vulnerability to future attacks. Indeed, the study by IBM Security reports that an overwhelming majority (83%) of the organizations responding to its survey experienced more than one data breach between March, 2021 and March 2022. Announcements of data breaches and their legal and financial ramifications are likely to raise investors' concerns and heighten crash risk. Moreover, modern information technologies tend to intensify market reactions to corporate information disclosures (Asthana & Balsam, 2001; Asthana, Balsam, & Sankaraguruswamy, 2004; Gao & Huang, 2020), which exacerbates crash risk. Given the possible opposing effects of the adoption of DBN laws on crash risk, their net effect is unclear ex ante and needs to be determined empirically.

We analyze a sample of 3,615 U.S. public firms over the period 1997–2019 and find that the enactment of DBN laws, which leads to an increase in data breach disclosures, is associated with higher future crash risk. Our evidence suggests a positive relation between data breach disclosures and crash risk. The effect is also economically meaningful. Our estimation suggests that the adoption of DBN laws is associated with an increase in crash risk that is equivalent to approximately 10% of its standard deviation.

The DiD approach is based on the parallel trend assumption that in the absence of the adoption of DBN laws, the crash risk of firms headquartered in states that do and those headquartered in states that do not adopt DBN laws would have evolved in a similar way. To examine the validity of this assumption, we estimate a crash risk dynamic model that tracks the evolvement of crash risk around the DBN adoption. We find that crash risk is not significantly different between the treatment and control firms in the years before the adoption of DBN laws, which substantiates the parallel trends assumption. Moreover, the treatment firms experience a significant increase in crash risk in the years after DBN laws adoption, suggesting that our finding is likely causal.

We run a battery of robustness tests to check the sensitivity of our results. To address concern that systematic differences between the treatment and control firms may drive our results, we employ the propensity score matching (PSM) and entropy balancing approaches for analysis but find qualitatively similar results. Furthermore, our results are robust to alternative crash risk measures.

Next, we examine whether the positive relation between the adoption of DBN laws and crash risk varies with firms' corporate governance. Previous studies document that self-interested managers are more likely to hide bad news over an extended period of time for personal gain (Kothari, Shu, & Wysocki, 2009), contributing to crash risk (Jin & Myers, 2006), whereas firms with strong corporate governance structure are associated with lower crash risk (An & Zhang, 2013; Andreou, Antoniou, Horton, & Louca, 2016; Callen & Fang, 2013; Chen, Kim, & Yao, 2017; Habib, Hasan, & Jiang, 2018). To the extent that managers of poorly governed firms are more likely to hide data breaches in the absence of DBN laws, the disclosure effect of DBN laws on crash risk should be more pronounced for poorly governed firms. Consistent with this expectation, we find that the effect of DBN laws on crash risk is stronger for firms with weaker corporate governance *ex ante*.

Financially constrained firms typically have limited access to the external capital markets (Korajczyk & Levy, 2003), therefore, these firms may underinvest in data security, which implies vulnerability to data breaches. As DBN laws mandate data breach disclosures and induce firms to invest in data security improvement (Romanosky et al., 2011), financially constrained firms may face difficulty in funding this type of investment, which suggests a stronger positive effect of DBN laws on crash risk for these firms. The cross-sectional analysis results are consistent with our prediction.

Information asymmetry can facilitate managers' bad news hoarding (Kothari et al., 2009). Therefore, we run a test to evaluate the effect of information asymmetry on the relation between DBN laws adoption and crash risk. Our results indicate that the positive impact of DBN laws on crash risk is more pronounced for firms with a higher degree of information asymmetry.

Finally, the adoption of DBN laws may increase other types of risk, such as downside risk (Ang, Chen, & Xing, 2006), tail risk (Kelly & Jiang, 2014), bear beta (Lu & Murray, 2019), left-tail momentum (Atilgan, Bali, Demirtas, & Gunaydin, 2020), crash sensitivity (Chabi-Yo, Huggenberger, & Weigert, 2022; Chabi-Yo, Ruenzi, & Weigert, 2018). To alleviate a concern that other types of risk explain our findings, we run an event study that examines the stock price reaction to data breach disclosures, particularly following the adoption of DBN laws. We find negative stock abnormal returns around data breach disclosures. Our evidence suggests that data breach disclosures are more likely to be associated with crash risk than other types of risk.

Our research contributes to literature in two important ways. First, among various factors that lead to stock price crashes, managers' withholding of bad news is a prominent one (Habib et al., 2018; He, 2015; Kim, Li, & Zhang, 2011a). The adoption of DBN laws curbs managers' ability to withhold information related to data breaches, leading to a sharp increase in the disclosures of data breaches, many of which accumulated over time. Our study shows that data breach disclosures, which have substantial legal and financial consequences for the breached firms and signal their data security vulnerability, are an important determinant of crash risk.[3] Second, our evidence suggests that the relation between data breach disclosures and crash risk varies with firm characteristics including corporate governance, financial constraints, and information asymmetry. A direct implication of our research findings for corporate managers and investors is that, while it is difficult to avoid sophisticated data breaches completely, firms can mitigate their harmful effect on crash risk by improving corporate governance to reduce managerial discretion in data breach disclosures, making investment to strengthen data security, and improving corporate information environment.

The rest of the paper is organized as follows. We provide institutional background and develop testable hypotheses in Section 2. Section 3 describes data and variables. Section 4 presents empirical results and discussions. Section 5 presents cross-sectional tests and Section 6 concludes.

[3] It came to our attention that Obaydin, Xu, and Zurbruegg (2021) were simultaneously working on a study similar to ours.

2. Institutional background and hypothesis development

2.1. Institutional background

"A data breach occurs when there is a loss or theft of, or other unauthorized access to, data containing sensitive personal information that results in the potential compromise of the confidentiality or integrity of data" (Stevens, 2012). Anecdotal evidence suggests that data breaches can harm the breached firms' reputation and deteriorate their financial conditions. For example, in 2017, Equifax, one of the three major US credit reporting agencies, announced a data breach in which cyberthieves extracted 605 million records of personal data including names, dates of birth, Social Security numbers, driver's license numbers, and credit card numbers of 147 million Americans. The Equifax case is unprecedented in terms of the breach scale and the sensitivity of information leaked, making it one of the most significant data breaches in history. Equifax stock price dropped 13% in early trading the day after the breach was disclosed, and a week later, the company announced the immediate departures and replacement of its Chief Information Officer and Chief Security Officer. The breach also triggered numerous lawsuits that cost the company $700 million in fines and compensation.[4] The Equifax case is only one of many recent data breaches disclosed that raise public concern about the negative effects of cybercrimes.

To strengthen customer protection and bring more attention to cybersecurity threats, regulators have enacted laws and standards on firms' cyber-related information disclosure for the past two decades. U.S. states gradually introduced formal data breach disclosure legislation beginning in the early 2000s. California was the first state to introduce such a legislation in 2002, requiring firms to disclose their data breaches. By late March 2018, all 50 states in the U.S. and the District of Columbia, had enacted DBN laws. Table 1 provides the timeline of DBN laws adoption over time. Because the residency of the affected customers rather than the location of the data breach drives the disclosure, a firm that experiences a data breach has to comply with the state law of each of its affected customers, which raises compliance challenges for firms that have operations in multiple states. For this reason, critics pointed out that DBN laws created "a fragmented, incoherent liability scheme" (Winn, 2009). Nevertheless, DBN laws proponents argue that "state laboratories can provide stronger protection for sensitive personal information."[5]

Table 1
DBN laws adoption by states.

State	Year	State	Year	State	Year
California	2002	Rhode Island	2005	Maryland	2007
Arkansas	2005	Tennessee	2005	Massachusetts	2007
Connecticut	2005	Washington	2005	Oregon	2007
Delaware	2005	Arizona	2006	Texas	2007
Florida	2005	Colorado	2006	Wyoming	2007
Georgia	2005	Hawaii	2006	Alaska	2008
Illinois	2005	Idaho	2006	Iowa	2008
Indiana	2005	Kansas	2006	Oklahoma	2008
Louisiana	2005	Michigan	2006	South Carolina	2008
Maine	2005	Montana	2006	Virginia	2008
Minnesota	2005	Nebraska	2006	West Virginia	2008
Nevada	2005	New Hampshire	2006	Missouri	2009
New Jersey	2005	Pennsylvania	2006	Mississippi	2010
New York	2005	Utah	2006	Kentucky	2014
North Carolina	2005	Vermont	2006	New Mexico	2017
North Dakota	2005	Wisconsin	2006	Alabama	2018
Ohio	2005	District of Columbia	2007	South Dakota	2018

This table presents the state-level adoption of the data breach notification laws for the period 2002–2018.

2.2. Hypothesis development

Before the adoption of DBN laws, U.S. firms were not obligated to disclose data breaches. DBN laws require businesses and entities affected by data breaches to promptly notify their customers and involved parties of their cyber incidents. These laws may reduce crash risk for the following reasons. First, DBN laws promote firm awareness of cybersecurity threats and create an incentive for firms, including those that have not experienced any data breaches, to invest in data security improvements, thereby "disinfecting themselves of shoddy security practices" (Ranger, 2007; Romanosky et al., 2011). This investment is expected to reduce the likelihood of breaches or at least mitigate any resulting harm from significant cyber incidents that could trigger extremely negative market reactions (Garg, Curtis, & Halper, 2003, Cavusoglu, Mishra, & Raghunathan, 2004; Garg, Curtis, & Halper, 2003; Gordon, Loeb, & Zhou, 2011; Lending et al., 2018; Tosun, 2021).

Second, DBN laws may encourage managers to adjust corporate disclosure behavior and accounting practices. It is difficult for investors to discover cyber incidents independently while self-interested managers have incentives to withhold negative information from investors to protect their reputation, job security, and compensation, thus, firms tend to underreport cyber incidents (Amir, Levi, & Livne, 2018). As DBN laws require firms to disclose cyber incidents, they curb managers' ability to hoard bad news, potentially lowering crash risk. Skinner (1994) finds that managers voluntarily disclose bad news early to avoid costs of potential litigation. Shroff, Venkataraman, & Zhang (2013) suggest that managers act more conservatively in response to material events (e.g., data breach) since litigation risk and costs are higher for material misstatements or omissions. Moreover, media coverage of data breaches attracts investors' attention to the breached firms' data security issues. Thus, media monitoring can play a complementary role in deterring firms from withholding bad news (An, Chen, Naiker, & Wang, 2020; Wen, Xu, Ouyang, & Kou, 2019 and Wu, Xiong, Gao, & Zhang, 2022).

Third, from the capital market perspective, a direct implication of data breach disclosures is lower information asymmetry among investors around cyber incidents. Hong & Stein (2003) develop a theory of market crashes based on differences of opinion among investors. They argue that in the presence of short-sale constraints, disagreement between informed and uninformed investors over a firm's fundamental value leads to more negative skewness in the distribution of stock returns, which implies higher crash risk. Chen, Hong, & Stein (2001) show empirically that investor disagreement is positively correlated with crash risk, which is consistent with Hong & Stein (2003) model. In a similar vein, Chang, Hsiao, Ljungqvist, & Tseng (2022) find that improved mandatory disclosure generally reduces information asymmetry between managers and outsiders, alleviating investor disagreement and decreasing crash risk. Taken together, the foregoing discussions suggest a negative relation between the adoption of DBN laws and crash risk.

A contrasting viewpoint posits that the implementation of DBN laws increases crash risk for at least two reasons. First, in the absence of mandatory data breach disclosure requirements, firms lack motivation to publicly announce incidents and tend to prioritize discreetly resolving data security issues to prevent customer and investor panic. Following DBN law adoption, managers must disclose both previously concealed and new data breaches, signifying past insecurity and susceptibility to

[4] https://www.cnbc.com/2019/07/22/what-you-need-to-know-equifax-data-breach-700-million-settlement.html

[5] U.S. Congress, House Committee on Energy and Commerce, Subcommittee on Commerce, Manufacturing, andTrade, *The Threat of Data Theft to American Consumers*, Hearing, 112th Cong., 1st sess., May 4, 2011, S. Hrg. 112–44 (Washington: GPO, 2011), p. 60.

future attacks. IBM Security's study underscores the inevitability of breaches, with most organizations experiencing multiple incidents within a year. The legal and financial ramifications of data breach disclosures attract investors' attention, sparking adverse market responses (Cavusoglu et al., 2004; Garg et al., 2003; Gordon et al., 2011; Lending et al., 2018; Tosun, 2021). Firms experiencing data breaches tend to face selling pressure and high trading volatility, resulting in extreme price movements (Goel & Shawky, 2009; Tosun, 2021). While the negative effects around the breach disclosure date are apparent, questions arise about their specific connection to crash risk or other factors, such as downside risk (Ang et al., 2006), bear beta (Lu & Murray, 2019), or crash sensitivity (Chabi-Yo et al., 2018; Chabi-Yo et al., 2022). As data breaches continue to grow in scale and severity, disclosures often trigger material valuation shocks over short time periods, elevating the likelihood of price crashes (Corbet & Gurdgiev, 2019).

Second, varying disclosure requirements across states grant firms operating in multiple jurisdictions flexibility in data breach notifications.[6] To mitigate possible consequences of data breach disclosures, managers may alter their disclosure behaviors, such as timing the disclosures, underreporting bad news related to data breaches (Amir et al., 2018; Foerderer & Schuetz, 2022; Freifeld, 2014; Tsukayama, 2017), and engaging in earnings manipulation (Xu et al., 2019), which can heighten future crash risk.[7] These discussions suggest that the adoption of DBN laws is positively related to crash risk.

Given the opposing arguments about the relation between data breach disclosures and crash risk, we need to determine such a relation empirically.

3. Sample, variable construction, and empirical model

3.1. Sample

We obtain accounting data from the Compustat and stock return data from the Centre for Research in Security Prices (CRSP) databases for the period 1997–2019. We then merge accounting and stock return data with analyst coverage data from I/B/E/S and institutional ownership data from Thomson Reuters 13F filings.

To examine the effect of DBN laws on crash risk, we construct the DBN indicator variable that equals 1 for the years in which DBN laws are effective in a firm's headquarters state, and 0 otherwise.[8] Our sample begins in 1997, five years before the first adoption of DBN law by California, and ends in 2019.[9] Furthermore, we restrict the sample period to 11 years centered on the DBN law passage year for each state to avoid possible confounding effects. Following previous studies (e.g., Chang, Chen, & Zolotoy, 2017; Hutton, Marcus, & Tehranian, 2009; Kim et al., 2011a; Kim, Li, Lu, & Yu, 2016; Kim, Li, & Zhang, 2011b), we exclude firms from the financial services and utilities industries (SIC 6000–6999 and SIC 4900–4999) and firm-year observations with missing financial and accounting data to calculate control variables in our analysis. The final sample includes 22,723 firm-year observations of 3615 unique U.S. public firms.

3.2. Measuring stock price crash risk

Following Chen et al. (2001) and Callen & Fang (2013), we calculate the negative skewness of stock returns, NCSKEW, as the primary measure of firm-specific crash risk. In particular, we first estimate the residuals from the expanded market model as follows:

$$R_{i,t} = \alpha_i + \beta_{1,i}R_{m,t-2} + \beta_{2,i}R_{m,t-1} + \beta_{3,i}R_{m,t} + \beta_{4,i}R_{m,t+1} + \beta_{5,i}R_{m,t+2} + \varepsilon_{i,t} \quad (1)$$

where $R_{i,t}$ is the daily stock return for firm i, α_i is the intercept, $R_{m,t}$ is the return of CRSP value-weighted market index, and $\varepsilon_{i,t}$ is the error term. We include two lead and two lag terms for daily market returns to account for nonsynchronous trading (Dimson, 1979). The firm-specific daily return ($D_{i,t}$) is calculated as the logarithm value of one plus the residual return from Eq. (1), i.e., $D_{i,t} = \ln(1 + \varepsilon_{i,t})$.

Negative conditional skewness, *NCSKEW*, is the negative of the third moment of each stock's firm-specific daily returns divided by the standard deviation of firm-specific daily returns raised to the third power. *NCSKEW* captures the negative asymmetry of the stock returns distribution and is calculated as:

$$NCSKEW_{i,t} = -\left[n(n-1)^{3/2}\sum D^3{}_{i,t}\right] \Big/ \left[(n-1)(n-2)\left(\sum D^2{}_{i,t}\right)\right]^{3/2} \quad (2)$$

where n is the number of observations of firm i daily returns during the fiscal year t. By construction, firms with higher *NCSKEW* are more likely to experience price crashes.

Our second measure of firm-specific crash risk is the natural logarithm of "down-to-up volatility," *DUVOL*, which is calculated as follows:

$$DUVOL_{i,t} = log\left\{\left[(n_u-1)\sum\nolimits_d D^2{}_{i,t}\right] \Big/ \left[(n_d-1)\right]\sum\nolimits_u D^2{}_{i,t}\right\} \quad (3)$$

where n_u and n_d are the number of up days and down days in fiscal year t, respectively. For each stock i over fiscal year t, we separate the days with returns below the period mean ("down" days) from those with returns above the period mean ("up" day) and compute the standard deviation for each of these groups separately. DUVOL is the natural logarithm of the ratio of the standard deviation on the down days to the standard deviation on the up days. A stock with a higher value of DUVOL is more prone to crash. Compared to NCSKEW, this alternative measure of crash risk is less likely to be biased by a handful of extreme returns as it does not involve third moments (Chen et al., 2001).

3.3. Control variables

We include a set of control variables that have been documented to affect crash risk (e.g., Chen et al., 2001; Hutton et al., 2009; Jin & Myers, 2006). Detrended turnover ($DTURN_{i,t}$) is the difference of average monthly share turnover between fiscal year t and t–1, where monthly share turnover is calculated as the monthly trading volume divided by the number of shares outstanding over the month. Stock return volatility ($SIGMA_{i,t}$) is the standard deviation of stock daily returns over the fiscal year t. Past stock returns ($RET_{i,t}$) is the average of stock daily returns over the fiscal year t. Chen et al. (2001) find that stocks with higher disagreement in investor opinions and past return mean and volatility are more prone to crash in the future. Firm size ($SIZE_{i,t}$) is the logarithm of the market value of equity at the end of fiscal year t. Market-to-book

[6] The threshold at which data breaches require notification is, depending on the state, between 1 and 1000 breached records. In addition, regulations differ in the time frames imposed on firms to notify affected consumers. Regulations urge firms to make data breach announcements after the first notice "in the most expedient time possible" (i.e., California Civil Code 1798.29(a)) or "as soon as practicable" (i.e., Massachusetts General Laws, Part I, Title XV, Chapter 93H, Section 3). Generally, firm must notice affected parties without reasonable delay, but no later than 60 days after discovering data breaches (Stevens, 2012).

[7] Anecdotal evidence shows that breached retailers (e.g., Target, Neiman Marcus) delay notifying their customers to maintain sales during the holiday shopping season (Freifeld, 2014).

[8] We determine a firm's state of headquarters based on Edgar electronic filings and data from Bill McDonald's website (https://sraf.nd.edu/data/)

[9] As we require five years before and five years after the adoption of DBN laws, we exclude three states that adopt DBN laws in or after 2017 (Alabama, New Mexico, and South Dakota) from the sample. However, including these states does not alter our conclusion. Also, as most states had adopted DBN laws by December 2008, we rerun our regressions with a sample period that ends in 2013. The results of this subsample are qualitatively similar to those of the full sample (the results are not reported but available upon request).

ratio ($MB_{i,t}$) is the market value of equity divided by the book value of equity at the end of fiscal year *t*. Financial leverage ($LEV_{i,t}$) is the book value of all liabilities scaled by total assets at the end of fiscal year *t*. Return on assets ($ROA_{i,t}$) is defined as income before extraordinary items divided by lagged total assets. Crash risk ($NCSKEW_{i,t}$) is the negative conditional skewness of firm-specific daily returns in fiscal year *t*. Following Hutton et al. (2009), we measure opacity ($OPAQUE_{i,t}$) as the absolute value of discretionary accruals averaged over the previous three years, where discretionary accruals are estimated by the modified Jones model (Dechow, Sloan, & Sweeney, 1995). We control for macroeconomic conditions of a state by including per capita income (PCI_t), gross state product (GSP_t), state-level unemployment ($UEMP_t$), and growth in gross state product (ΔGSP_t). All continuous variables are winsorized at the 1% and 99% levels to limit possible effects of outliers. We provide the definitions of the variables in Appendix 1.

3.4. Empirical model

We employ the DiD approach to examine the effect of DBN laws on crash risk. Our treatment (control) group includes firms headquartered in states that have (have not) adopted the DBN laws. Specifically, our baseline regression model has the following form:

$$CRASH\ RISK_{i,t+1} = \beta_0 + \beta_1 DBN_{i,t} + \boldsymbol{\delta X_{i,t}} + Firm\ fixed\ effects + Year\ fixed\ effect + \varepsilon_{i,t} \quad (4)$$

where *i* and *t* represent firm and year, respectively. The dependent variable, $CRASH\ RISK_{t+1}$, is either $NCSKEW_{t+1}$ or $DUVOL_{t+1}$. The variable of interest is the indicator variable $DBN_{i,t}$. If the passage of DBN laws increases crash risk, the coefficient β_1 should be positive and statistically significant. By contrast, if the adoption of DBN laws reduces crash risk, β_1 should be negative and statistically significant. $\boldsymbol{X_{i,t}}$ is a vector of control variables.

Table 2
Summary statistics.

Variable	N	Mean	Std. Dev.	p25	Median	p75
Crash Risk Variables						
NCSKEW	22,723	0.014	1.345	−0.592	−0.117	0.381
DUVOL	22,723	−0.066	0.302	−0.254	−0.081	0.097
Indicator Variable						
DBN	22,723	0.407	0.491	0	0	1
Firm-level Variables						
DTURN	22,723	0.002	0.112	−0.033	0	0.034
RET (%)	22,723	−0.086	0.115	−0.102	−0.045	−0.021
SIGMA	22,723	0.036	0.021	0.021	0.03	0.045
SIZE	22,723	5.74	2.108	4.227	5.747	7.116
MB	22,723	2.903	4.615	1.168	2.003	3.522
LEV	22,723	0.19	0.21	0.004	0.134	0.302
ROA	22,723	−0.023	0.227	−0.051	0.034	0.087
OPAQUE	22,723	0.872	1.377	0.184	0.393	0.888
State-level Variables						
PCI ($ mil.)	22,723	0.063	0.018	0.054	0.064	0.079
GSP ($ mil.)	22,723	1.153	1.008	0.380	0.630	1.795
UEMP (%)	22,723	4.433	1.569	3.5	4	4.7
ΔGSP (%)	22,723	4.586	1.687	3.624	4.702	5.459

This table reports the descriptive statistics for stock price crash risk and control variables. The sample includes 22,723 firm-year observations of 3,615 unique U. S. public firms between 1997 and 2019. All continuous variables are winsorized at the 1% and 99% levels. The definition of the variables is presented in Appendix 1.

4. Empirical results and discussions

4.1. Summary statistics and baseline regressions

Table 2 reports the summary statistics of the sample. *DBN* indicator has a mean of 0.407, suggesting that 40.7% of firm-year observations are affected by DBN laws during the sample period. The summary statistics of the control variables are largely in line with those documented in the literature.

Table 3 reports the baseline regression results with heteroscedasticity-robust standard errors clustered by headquarters states. In Columns 1 and 6, we regress two crash risk measures, $NCSKEW_{t+1}$ and $DUVOL_{t+1}$, on the indicator variable, DBN_t, and firm and year fixed effects. We find that the coefficient on *DBN* is positive (0.11 and 0.02 in Columns 1 and 6, respectively) and statistically significant at the 1% and 5% levels. We include the set of control variables in the remaining columns. The coefficient on *DBN* remains positive and statistically significant for both crash risk measures. The effect of DBN laws is also economically meaningful. For example, the coefficient estimates (0.142 and 0.029) of *DBN* in Columns 2 and 7 of Table 3 indicate that the adoption of DBN laws leads to an increase in crash risk of the treatment firms that is equivalent to 10.56% (=0.142/1.345) and 9.6% (=0.029/0.302) of the standard deviations of *NCSKEW* and *DUVOL*, respectively. This finding suggests that the adoption of DBN laws heightens a firm's crash risk. We further find that the coefficient of detrended share turnover ($DTURN_{i,t}$) is positive and statistically significant, suggesting that stocks with higher turnover are more prone to crash risk (Chen et al., 2001). The coefficients on other control variables such as $SIZE_{i,t}$, $LEV_{i,t}$, $MB_{i,t}$, $ROA_{i,t}$, and $NCSKEW_{i,t}$ are largely in line with those documented in the prior studies (e.g., Callen & Fang, 2013; Hutton et al., 2009).

Crash risk can be correlated with unobserved firm characteristics, state business environment, industry common factors, and time-varying macroeconomic factors, therefore, we additionally control for different sets of fixed effects including firm and year fixed effects, industry-by-year fixed effects, and state fixed effects. The results, reported in Columns 3–5 and 8–10 of Table 3, indicate that our finding continues to hold.

4.2. Dynamic model, PSM, and robustness checks

4.2.1. Dynamic model

Our identification is grounded on the argument that the staggered adoption of DBN laws represents a plausibly exogenous shock to managers' ability to hoard data breach information, thereby affecting the treatment firms' crash risk.[10] However, it is possible that firms' crash risk and the adoption of DBN laws by states simply follow time trends, which implies that the relation between DBN laws and crash risk is spurious rather than causal. To address this concern, we follow Bertrand & Mullainathan (2003) and Cornaggia, Mao, Tian, & Wolfe (2015) and investigate the dynamic effects of DBN law adoption on crash risk. If pre-treatment trends drive the results, we expect to observe significant effects of DBN laws on firm's crash risk prior to their passage.

We employ a dynamic model to test whether firms headquartered in

[10] Because states choose to adopt DBN laws, such an adoption is not random (e.g., data breach incidents that occurred in a state drive the enactment of state's DBN law). Therefore, we use the Heckman's selection model to address self-selection bias concern. Specifically, in the first stage, we estimate the probability that a state adopts DBN law in a given year by using a probit model with the DBN laws adoption dummy as the dependent variable and the total number of data breaches occurred in a state and other state-level variables as controls. The predicted probability of DBN law adoption is then included in the crash risk equation in the second stage. Our finding is qualitatively unchanged (the results are not reported for brevity but available from the authors).

Table 3
DBN laws and crash risk - baseline regressions.

Variable	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
DBN_t	0.110***	0.142***	0.098***	0.113***	0.149***	0.021**	0.029***	0.019***	0.023***	0.030***
	(2.86)	(3.65)	(3.13)	(3.01)	(3.72)	(2.55)	(3.54)	(2.83)	(2.77)	(3.32)
$NCSKEW_t$		−0.098***	0.034***	0.035***	−0.100***		−0.019***	0.008***	0.008***	−0.019***
		(5.87)	(2.79)	(2.95)	(5.55)		(5.13)	(3.42)	(3.59)	(4.87)
$DTURN_t$		0.390***	0.503***	0.496***	0.366***		0.070***	0.112***	0.113***	0.061***
		(4.82)	(7.42)	(7.19)	(4.36)		(3.54)	(6.53)	(6.83)	(3.04)
RET_t		−0.444	1.372***	1.463***	−0.541		0.019	0.253***	0.254***	0.022
		(1.18)	(5.71)	(5.84)	(1.54)		(0.23)	(4.84)	(4.70)	(0.27)
$SIGMA_t$		−5.292*	6.032***	6.851***	−4.814*		−0.718	0.416	0.418	−0.370
		(2.00)	(3.17)	(3.48)	(1.89)		(1.37)	(1.13)	(1.04)	(0.68)
$SIZE_t$		0.442***	0.093***	0.093***	0.472***		0.113***	0.021***	0.020***	0.121***
		(18.85)	(14.93)	(14.34)	(19.50)		(21.58)	(16.08)	(17.64)	(23.75)
MB_t		0.001	0.004**	0.005***	0.002		0.000	0.001***	0.001***	0.001
		(0.55)	(2.59)	(3.11)	(1.10)		(0.82)	(3.08)	(3.47)	(1.34)
LEV_t		0.325***	−0.071	−0.126**	0.340***		0.060***	−0.024**	−0.028**	0.061***
		(3.49)	(1.59)	(2.35)	(3.64)		(3.17)	(2.09)	(2.09)	(3.04)
ROA_t		0.132**	0.294***	0.262***	0.130***		0.052***	0.108***	0.103***	0.048***
		(2.68)	(7.38)	(6.61)	(2.70)		(4.28)	(9.31)	(9.29)	(4.17)
$OPAQUE_t$		−0.005	−0.007	0.004	−0.008		−0.002	−0.002*	−0.001	−0.003
		(0.50)	(1.02)	(0.63)	(0.58)		(0.80)	(1.69)	(0.78)	(0.97)
PCI_t		0.565	−0.597	0.351	0.615		0.448	−0.319**	−0.076	0.471
		(0.17)	(0.96)	(0.32)	(0.17)		(0.65)	(2.43)	(0.29)	(0.62)
GSP_t		−0.000	−0.000	−0.000	−0.000		−0.000	0.000	0.000	−0.000
		(0.92)	(0.17)	(0.19)	(0.81)		(1.14)	(0.70)	(0.85)	(1.02)
$UEMP_t$		−0.008	−0.013	−0.007	−0.007		−0.001	−0.004**	−0.002	−0.001
		(0.63)	(1.47)	(0.62)	(0.52)		(0.50)	(2.47)	(0.91)	(0.34)
ΔGSP_t		−0.013	−0.010*	−0.009	−0.011		−0.003	−0.003**	−0.002	−0.002
		(1.14)	(1.79)	(1.18)	(0.97)		(1.26)	(2.57)	(1.38)	(0.92)
Firm FE	Yes	Yes	No	No	Yes	Yes	Yes	No	No	Yes
Industry FE	No	No	Yes	No	No	No	No	Yes	No	No
State FE	No	No	No	Yes	No	No	No	No	Yes	No
Year FE	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No
Industry-year FE	No	No	No	No	Yes	No	No	No	No	Yes
No. of obs.	22,723	22,723	22,723	22,723	22,723	22,723	22,723	22,723	22,723	22,723
Adjusted R^2	0.05	0.11	0.04	0.04	0.11	0.07	0.14	0.07	0.07	0.15

This table presents the results of the crash risk regressions. The dependent variable, crash risk, is measured as negative conditional skewness (NCSKEW) or down-to-up volatility (DUVOL) in year t + 1. DBN indicator is a dummy variable that equals one for the years in which DBN law is effective in a firm's state of headquarters, and zero otherwise. Other variables are defined in Appendix 1. The sample consists of 22,723 firm-year observations for the period 1997 to 2019. The models are estimated with firm or industry fixed effect and year and state fixed effects or industry-year fixed effects. *t*-statistics based on heteroskedasticity-robust standard errors clustered by headquarters states are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

states that pass DBN laws (i.e., the treatment group) and those that have not passed them (i.e., the control group) satisfy the pre-treatment parallel trends assumption. Similar to Cornaggia et al. (2015), we construct four dummy variables indicating four periods around the adoption of DBN event: $Before^{2+}$ is a dummy variable that equals 1 for two or more years preceding the adoption of DBN laws by the state of headquarters of a given firm, and 0 otherwise. $Before^{1}$ is a dummy variable that equals 1 for the year preceding DBN laws adoption, and 0 otherwise. $After^{1}$ is a dummy variable that equals 1 for the first year after the passage of DBN laws, and 0 otherwise. $After^{2+}$ is a dummy variable that equals 1 for two or more years after the passage of DBN laws, and 0 otherwise. By construction, the year in which a DBN law was adopted is the base year. The dynamic model has the following form:

$$Crash\ Risk_{i,t+1} = \beta_0 + \beta_1 Before^{2+}{}_{i,t} + \beta_2 Before^{1}{}_{i,t} + \beta_3 After^{1}{}_{i,t} + \beta_4 After^{2+}{}_{i,t} + \boldsymbol{\delta X_{i,t}} + Firm\ fixed\ effects + Year\ fixed\ effects + \varepsilon_{i,t} \quad (5)$$

We regress both measures of crash risk on the four period indicators while controlling for other firm and state characteristics, firm fixed effects, and year and state or industry-by-year fixed effects. The results reported in Table 4 indicate that the coefficients on $Before^{2+}$ and $Before^{1}$ are statistically insignificant, whereas the coefficients on $After^{1}$ and $After^{2+}$ are positive and statistically significant.

Similar to Beck, Levine, & Levkov (2010) and Dang, Lee, Liu, & Zeng (2018), in a complementary analysis we examine the effect of DBN laws on crash risk in each year around their adoption years and plot the coefficient estimates in Fig. 1. We find that the positive effect of DBN laws on crash risk is statistically significant following their adoption but not beforehand. Collectively, the results substantiate the pre-trend parallel assumption underlying the DiD model and suggest that our finding is likely causal.

4.2.2. PSM analysis

Our results thus far suggest a positive relation between DBN laws adoption and crash risk. In this section, we use the PSM approach to address a concern that firms headquartered in states that adopt DBN laws are fundamentally different from the remaining firms and such differences drive our results.

An empirical challenge in creating a propensity score-matched sample in our setting is that the adoption of DBN laws is staggered over time. In particular, firms in the control group at the beginning of the sample could be in the treatment group near the end of the sample. To address this concern, we follow previous studies (e.g., Appel, 2019; Gormley & Matsa, 2011; Ni & Yin, 2018) and construct a cohort of control and treatment firms using firm-year observations for the three years before and the three years after each adoption of DBN laws. Within each cohort, we run a logistic regression to estimate propensity score for each firm based on all control variables included in the baseline model (Eq. 4). We then match each treated firm (i.e., firm headquartered in state that adopted DBN laws) with a control firm (i.e., firms headquartered in states that had not adopted DBN laws) on the nearest propensity score without replacement. We repeat this process for each

Table 4
DBN laws and crash risk - dynamic model.

Variable	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)
$Before^{2+}$	−0.041	−0.018	−0.031	−0.011	−0.006	−0.009
	(1.12)	(0.56)	(0.77)	(1.27)	(0.76)	(1.07)
$Before^{1}$	−0.016	0.021	−0.006	−0.002	0.007	0.002
	(0.39)	(0.51)	(0.14)	(0.20)	(0.78)	(0.17)
$After^{1}$	0.130***	0.129***	0.151***	0.026***	0.027**	0.031***
	(3.19)	(2.78)	(3.54)	(2.72)	(2.51)	(2.96)
$After^{2+}$	0.161***	0.117**	0.152***	0.037***	0.027***	0.035***
	(3.59)	(2.62)	(3.43)	(4.15)	(2.90)	(3.88)
Intercept	−2.376***	−0.636***	−2.598***	−0.702***	−0.164***	−0.767***
	(8.73)	(4.31)	(8.37)	(12.69)	(5.99)	(12.13)
Firm & State Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	No	Yes	Yes	No	Yes
Industry FE	No	No	No	No	No	No
State FE	No	Yes	No	No	Yes	No
Year FE	Yes	Yes	No	Yes	Yes	No
Industry-year FE	No	No	Yes	No	No	Yes
No. of obs.	22,723	22,723	22,723	22,723	22,723	22,723
Adjusted R^2	0.11	0.04	0.11	0.14	0.07	0.15

This table presents the estimation results of the dynamic crash risk model. The dependent variable, crash risk, is measured as negative conditional skewness (NCSKEW) or down-to-up volatility (DUVOL) in year t + 1. *Before*^{1} (*Before*$^{2+}$) is a dummy variable that equals one for observations with one (two or more) year(s) prior to DBN laws adoption and zero otherwise. *After*^{1} (*After*$^{2+}$) is an indicator variable that equals one for observations with one (two or more) year(s) after the DBN laws adoption, and zero otherwise. Other variables are defined in Appendix 1. The models are estimated with firm fixed effects, and year and industry-year fixed effects or state-year fixed effects, but their estimates are suppressed for brevity. *t*-statistics based on heteroskedasticity-robust standard errors clustered by headquarters states are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

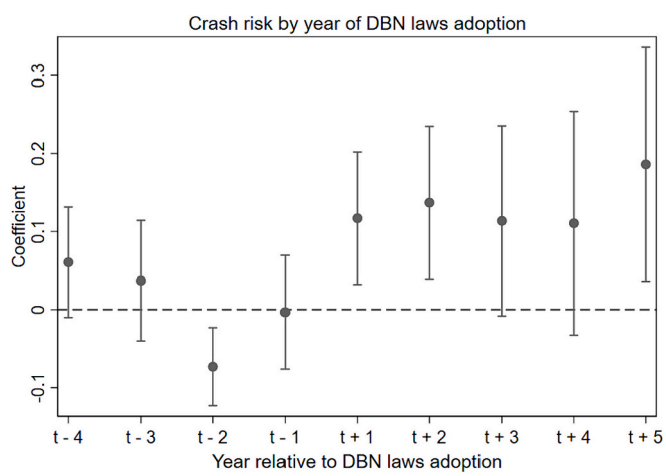

Fig. 1. Crash Risk by Year of DBN Laws Adoption.
This figure shows the dynamic effects of DBN laws adoption on crash risk. Crash risk is measured as negative conditional skewness (NCSKEW). We estimate the following specification including lead and lag indicators of DBN laws adoption: $\text{Crash Risk}_{i,t+1} = \alpha + \beta_1 D_{i,t-4} + \beta_2 D_{i,t-3} + \ldots + \beta_9 D_{i,t+5} + \text{Controls} + \text{Year}_t + \text{State}_i + \varepsilon_{i,t}$ where $D_{i,t}$ is a dummy variable set to one if a state adopts DBN laws in year t, and zero otherwise. $D_{i,t-\alpha}$ is set to one for the period up to α years prior to DBN laws adoption, and zero otherwise. $D_{i,t+\alpha}$ is set to one for the period α years after DBN laws adoption, and zero otherwise. We report the estimated coefficients with 95% confidence intervals, adjusted for state-level clustering.

DBN laws cohort and stack all cohorts together to form our final PSM sample.

Panels A and B of Table 5 report a univariate comparison of the treatment and control firms' characteristics before and after the match, respectively. The differences in post-match firm characteristics are not statistically significant between the two groups, implying a successful match. Panel C of Table 5 reports the crash risk regressions using the propensity score-matched sample. Consistent with our earlier finding, the coefficients on the DBN laws indicator are all positive and significant. Overall, the results from the PSM analysis lend further support to our finding.

A concern with the PSM approach is that it may not directly focus on producing covariate balance (Hainmueller, 2012). In particular, matching may improve balance on some covariates while decreasing balance on other covariates, thereby counteracting bias reduction for the subsequent treatment effect estimation (e.g., Diamond & Sekhon, 2013; Ho, Imai, King, & Stuart, 2007; Iacus, King, & Porro, 2012). To address this concern, we use the entropy balancing approach (Hainmueller & Xu, 2013) and rerun our baseline regressions. This approach assigns continuous weights to control firms while retaining the full sample so that the means and variances of the observable characteristics of the treated and control firms are statistically indifferent. The results presented in Table A1 in the Online Appendix suggest that our findings are essentially unchanged.

4.2.3. Robustness checks

We run a battery of robustness checks to examine the sensitivity of our results. Specifically, we employ three other measures of crash risk including *CRASH*, an indicator variable that equals 1 if a firm experiences one or more daily returns falling 3.09 or more standard deviations below the mean daily return during its fiscal year, and 0 otherwise (Hutton et al., 2009; Kim et al., 2011a, 2011b); *CRASH_COUNT*, measured as the difference in the number of days with negative extreme firm-specific daily returns and the number of days with positive extreme firm-specific daily returns (Callen & Fang, 2015a, 2015b)[11]; and *EXTRASIGMA*, measured as the negative of the worst deviation of firm-specific daily returns from the average firm-specific daily return divided by the standard deviation of firm-specific daily returns (Andreou et al., 2016; Ni, Wang, & Yin, 2021).[12] Columns 1–3 of Panel A in Table 6 present results of these alternative crash risk measures regressions. The coefficients on the DBN laws indicator variable remain positive and significant, suggesting that our findings are robust to alternative

[11] Extreme daily returns are defined as daily returns exceeding 3.09 standard deviations below (above) the mean daily return over the fiscal year.

[12] Following Andreou et al. (2016) and Ni et al. (2021), we calculate the extreme sigma as $\text{EXTRASIGMA} = -\text{Min}\left[\frac{D_{i,t}-D}{\sigma D}\right]$ where $D_{i,t}$ is the firm-specific daily return, D is the daily mean return, and σ_D is the standard deviation of daily return over the fiscal year.

Table 5
DBN laws and crash risk - propensity score matching.

Panel A: Differences in Characteristics of Treatment and Control Firms Before Match

Variable	Control	Treatment	Difference	t-statistics
NCSKEW	−0.014	0.068	−0.082	−6.09
DUVOL	−0.077	−0.041	−0.036	−12.1
DTURN	0.002	0.007	−0.005	−0.49
RET	−0.123	−0.055	−0.068	−60.19
SIGMA	0.043	0.028	0.015	73.54
SIZE	5.2	6.4	−1.18	−56.93
MB	2.86	3.09	−0.23	−4.88
LEV	0.22	0.21	0.01	5.07
ROA	−0.04	−0.12	−0.03	−14.77
OPAQUE	0.67	0.9	−0.23	−20.42

Panel B: Differences in Characteristics of Treatment and Control Firms Post Match

Variable	Control	Treatment	Difference	t-statistics
NCSKEW	0.047	0.036	0.011	0.41
DUVOL	−0.056	−0.059	0.002	0.38
DTURN	0.008	0.009	−0.001	−0.63
RET	−0.071	−0.07	−0.001	−0.29
SIGMA	0.033	0.033	0	0.22
SIZE	5.825	5.851	−0.026	−0.63
MB	2.872	2.87	0.003	0.04
LEV	0.195	0.192	0.003	0.87
ROA	−0.018	−0.017	−0.001	−0.17
OPAQUE	0.961	0.956	0.004	0.14

Panel C. Crash Risk Regressions Using Propensity Score-Matched Sample

	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
DBN_t	0.163**	0.151**	0.117***	0.154**	0.159**	0.038**	0.035**	0.024**	0.031**	0.035**
	(2.30)	(2.33)	(2.72)	(2.50)	(2.33)	(2.20)	(2.20)	(2.44)	(2.07)	(2.20)
Firm & State Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	No	No	Yes	Yes	Yes	No	No	Yes
Industry FE	No	No	Yes	No	No	No	No	Yes	No	No
State FE	No	No	No	Yes	No	No	No	No	Yes	No
Year FE	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No
Industry-year FE	No	No	No	No	Yes	No	No	No	No	Yes
No. of obs.	9629	9629	9629	9629	9629	9629	9629	9629	9629	9629
Adjusted R^2	0.06	0.13	0.04	0.04	0.12	0.07	0.16	0.06	0.06	0.16

This table reports the crash risk regression results using the propensity score matched sample. Firms are matched on all baseline control variables. Panel A reports the differences in characteristics of treatment and control firms before matching. Panel B reports the differences in characteristics of treatment and control firms after matching. Panel C reports results of the crash risk regressions using the propensity score-matched sample. All variables are defined in Appendix 1. The models are estimated with firm- and year- or firm- and industry-year fixed effects, but their estimates are suppressed for brevity. *t*-statistics based on heteroskedasticity-robust standard errors clustered by states of headquarters are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

measures of crash risk.

Firms headquartered in states that adopted DBN laws later could observe the effects of DBN laws on businesses in states that had adopted the laws earlier, therefore, they would change corporate policies accordingly without waiting for these laws to be adopted by their own states. This possibility raises concerns about the exogeneity and actual effects of the laws, particularly for firms in late-adopter states. To address this concern, we run analysis for the subperiod 2000–2004, which includes firms in California, which adopted the DBN law in 2002, as treatment firms. The results reported in Column 1–2 of Panel B in Table 6 suggest that our results are qualitative unchanged. In a complementary analysis, we rerun the crash risk regressions for a subsample of firms headquartered in states that adopted DBN laws early in the sample period, from 2002 to 2006. The results reported in Table A2 in the Online Appendix suggest that our results are qualitative unchanged.

Our sample includes 930 firms headquartered in California, which accounts for 25.73% of the number of firms in the sample. To address a concern that our results could be biased by a large proportion of firms headquartered in California, we exclude these firms and rerun the crash risk regression. The results reported in Column 3–4 of Panel B in Table 6 indicate that our results continue to hold.

Previous research documents that firms in high litigation industries, including biotechnology, computing, electronics, and retailing, follow a different pattern of stock price crashes (Fang, Liu, & Xin, 2010; Francis, Philbrick, & Schipper, 1994).[13] Therefore, in another robustness check, we rerun the crash risk regression while excluding firms in high litigation industries. The results reported in Column 5–6 of Panel B in Table 6 indicate that the coefficient on DBN law indicator remains positive and statistically significant, suggesting that the effect of DBN laws is not sensitive to excluding firms in high litigation industries.

[13] Francis et al. (1994) suggests that managers of firms in high litigation industries have an incentive to voluntarily and early disclose bad news to reduce both the probability of being sued and the magnitude of estimated damages if there is a suit.

Table 6
Additional robustness tests.

Panel A: Alternative Measures of Crash Risk

Variable	$CRASH_COUNT_{t+1}$	$CRASHt+1$	$EXTRASIGMA_{t+1}$
	(1)	(2)	(3)
DBN_t	0.129***	0.026**	0.181***
	(2.91)	(2.54)	(3.31)
Firm & State Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
No. of obs.	22,267	22,267	22,267
Adjusted R^2	0.09	0.05	0.14

Panel B: Sensitivity Checks

Variable	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Subperiod 2000–2004		Excluding California Firms		Excluding Firms in High Litigation Industries		Excluding Global Crisis Period 2008–2009	
DBN_t	0.285***	0.060***	0.102*	0.022**	0.122***	0.021**	0.183***	0.036***
	(6.24)	(5.69)	(1.96)	(2.04)	(2.70)	(2.24)	(3.78)	(3.29)
Firm & State Controls	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	10,130	10,130	17,588	17,588	14,570	14,570	17,213	17,213
Adjusted R^2	0.16	0.21	0.1	0.13	0.1	0.13	0.11	0.15

This table reports the results of additional robustness tests. Columns 1–3 of Panel A report the regression results with alternative measures of crash risk as the dependent variable. Panel B presents results of sensitivity checks. Column 1–2 of Panel B reports results of the regression that includes firms in state (California) that adopted DBN laws between 2002 and 2004. Column 3–4 of Panel B reports results of the regression that excludes firms headquartered in California. In Column 5–6 of Panel B, we rerun the regression for a subsample that excludes high litigation industries (i.e., those from biotechnology, computing, electronics, and retailing industries). Column 7–8 of Panel B reports results of the regression that excludes the 2007–2008 financial crisis period. See Appendix 1 for variable definitions. *t*-statistics based on heteroskedasticity-robust standard errors clustered by firm headquarter state are reported in parentheses. Coefficients marked with *, **, and *** are significant at 10%, 5%, and 1%, respectively.

Table 7
Crash risk, DBN laws, and geographic dispersion.

Variable	$NCSKEW_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)
DBN_t	0.094*	0.015
	(1.83)	(1.32)
$LOW_GEO_DISP_t$	0.001	−0.001
	(0.01)	(0.7)
$DBN*LOW_GEO_DISP_t$	0.142***	0.041***
	(2.81)	(3.18)
Firm & State Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
No. of obs.	17,098	17,098
Adjusted R^2	0.12	0.16

This table reports results of crash risk regressions conditioning on the firm's geographic dispersion. The dependent variable is crash risk, measured by NCSKEW and DUVOL. Geographic dispersion is the number of states in which a firm does business. We set a dummy variable LOW_GEO_DISP equal to one for firms with geographic dispersion below the sample median, and zero otherwise. Other variables are as defined in Appendix 1. All models are estimated with firm- and year- fixed effects. *t*-statistics based on heteroskedasticity-robust standard errors clustered by state of headquarters are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8
DBN laws, corporate governance, and crash risk.

Variable	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)	(3)	(4)
DBN_t	0.052	0.050	0.000	0.002
	(1.08)	(1.04)	(0.03)	(0.25)
$DINST_t$	−0.200***		−0.058***	
	(5.19)		(6.71)	
$DBN*DINST_t$	0.196***		0.062***	
	(4.12)		(7.86)	
$DANA_t$		−0.154***		−0.038***
		(3.93)		(4.34)
$DBN*DANA_t$		0.190***		0.056***
		(3.99)		(6.09)
Firm & State Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
No. of obs.	22,267	22,267	22,267	22,267
Adjusted R^2	0.11	0.11	0.15	0.14

This table reports results of cross-sectional regressions. The dependent variable is crash risk, measured by NCSKEW and DUVOL. We set a dummy variable DINST (DANA) equal to one for firms with institutional ownership (analyst coverage) below its sample median, and zero otherwise. The higher value of the dummy variables indicates a weaker corporate governance. Other variables are as defined in Appendix 1. All models are estimated with firm- and year- fixed effects. *t*-statistics based on heteroskedasticity-robust standard errors clustered by state of headquarters are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Our sample period includes the Great Recession period 2007–2008, which exposed firms to higher crash risk. To mitigate possible confounding effects of the Great Recession, we drop the subperiod 2007–2008 from the sample and rerun crash risk regressions. The results reported in Columns 7–8 of Panel B in Table 6 suggest that our finding persists.

Lastly, because customer residency affects data breach disclosures, one may argue that firms' level of compliance with DBN laws depends on their geographical operation, with stronger effects of DBN laws on crash risk expected for firms with more operation concentrated in their headquarters states. To examine this possibility, we rerun analysis for

Table 9
DBN laws, financial constraints, and crash risk.

Panel A: Subsample Analysis.

Variable	$NCSKEW_{t+1}$		$DUVOL_{t+1}$		$NCSKEW_{t+1}$		$DUVOL_{t+1}$	
	Small Size	Big Size	Small Size	Big Size	High SA_Index	Low SA_Index	High SA_Index	Low SA_Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DBN_t	0.173***	0.075	0.042***	0.011	0.199***	0.094*	0.043***	0.016
	(3.17)	(1.41)	(3.30)	(1.16)	(3.58)	(1.80)	(3.36)	(1.60)
Firm & State Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of obs.	10,781	11,033	10,781	11,033	10,886	11,166	10,886	11,166
Adjusted R^2	0.12	0.1	0.17	0.11	0.15	0.08	0.19	0.1

Panel B: Triple Difference Test.

Variable	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)	(3)	(4)
DBN_t	0.038	0.046	−0.005	−0.001
	(0.82)	(0.93)	(0.62)	(0.17)
$DSIZE_t$	−0.195***		−0.047***	
	(3.26)		(3.51)	
$DBN*DSIZE_t$	0.236***		0.076***	
	(4.40)		(7.51)	
DSA_INDEX_t		−0.024		−0.010
		(0.41)		(0.90)
$DBN*DSA_INDEX_t$		0.198***		0.063***
		(3.48)		(5.77)
Firm & State Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
No. of obs.	22,267	22,267	22,267	22,267
Adjusted R^2	0.11	0.11	0.15	0.14

This table reports results of the crash risk regressions conditional on financial constraints. The levels of financial constraints are determined based on firm size or the SA index. In Panel A, Column 1 (Column 2) and Column 3 (Column 4) report results for firms with size below (above) the sample median. Column 5 (Column 6) and Column 7 (Column 8) report results for firms with SA_Index above (below) the sample median. In Columns 1 and 3 of Panel B, we set the DSIZE dummy variable equal to one for firms with firm size below the sample median, and zero otherwise. In Columns 2 and 4 of Panel B, we set the DSA_INDEX dummy variable equal to one for firms with SA_INDEX above its sample median, and zero otherwise. DBN indicator is a dummy variable that equals one for the years in which DBN law is effective in a firm's state of headquarters, and zero otherwise. Other variables are defined in Appendix 1. The models are estimated with firm and year fixed effects, but their estimates are suppressed for brevity. *t*-statistics based on heteroskedasticity-robust standard errors clustered by headquarters state are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

subgroups of firms sorted on their geographical dispersion. Following Garcia & Norli (2012), we measure a firm's geographic dispersion as the number of states in which the firm does business, with a larger number implying more dispersion. We run triple-difference regressions where we interact the DBN laws indicator variable with an indicator variable capturing the degree of firms' geographic dispersion of operation. Consistent with our expectation, the results presented in Table 7 indicate that the positive effect of DBN laws on crash risk is stronger for firms with lower geographical operation dispersion. In an unreported analysis, we exclude firms from the retail and wholesale industries, which typically have customers in multiple states, from the analysis but our results continue to hold.

5. Cross-sectional analysis

5.1. Corporate governance and crash risk

Kim et al. (2011b) and Callen & Fang (2013) document that stock price crashes are more likely for firms with managerial agency problems. Hermalin & Weisbach (2012) argue that a firm's disclosure policy is associated with its corporate governance quality. Managers of poorly governed firms are less likely to be held accountable for not releasing high-quality information in a timely manner (Bae, Lim, & Wei, 2006; Bhojraj & Sengupta, 2003). Therefore, if managers' bad news withholding is the driver of the relation between DBN laws adoption and crash risk, we expect such a relation to be more pronounced for firms with weaker corporate governance. To examine whether the relation between DBN laws adoption and crash risk varies with corporate governance, we perform triple-difference regressions where we interact the DBN laws indicator variable with the proxies for corporate governance.

We employ two proxies for corporate governance including institutional ownership (An & Zhang, 2013; Mitra & Cready, 2005; Velury & Jenkins, 2006) and analyst coverage (Chowdhury, Faff, & Hoang, 2021; Kim, Lu, & Yu, 2019; Piotroski & Roulstone, 2004). Callen & Fang (2013) show that institutional ownership is negatively associated with future crash risk, supporting the monitoring role of institutional investors. Previous studies argue that financial analysts play an important governance role in scrutinizing management behavior (e.g., Chen, Harford, & Lin, 2015; Yu, 2008). This view suggests that greater analyst coverage can reduce crash risk by curbing managers' ability to withhold bad news. We define the dummy variable DINST (DANA) as 1 for firms with below sample median values for institutional ownership (analyst coverage), and 0 otherwise. A higher value of each dummy variable indicates weaker corporate governance.

We run regressions of crash risk on DB laws indicator, corporate governance measures, and their interactions while controlling for other variables. The results reported in Table 8 indicate that the coefficients of

Table 10
DBN laws, information asymmetry, and crash risk.

Variable	$NCSKEW_{t+1}$	$NCSKEW_{t+1}$	$DUVOL_{t+1}$	$DUVOL_{t+1}$
	(1)	(2)	(3)	(4)
DBN_t	−0.009	0.052	−0.007	0.010
	(0.17)	(1.03)	(0.72)	(0.91)
DES_t	−0.107*		−0.010	
	(1.69)		(0.84)	
$DBN*DES_t$	0.243***		0.058***	
	(4.46)		(5.69)	
DQS_t		−0.016		0.009
		(0.33)		(0.86)
$DBN*DQS_t$		6.168***		1.475***
		(3.81)		(4.86)
Firm & State Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
No. of obs.	22,267	12,749	22,267	12,749
Adjusted R^2	0.11	0.11	0.14	0.13

This table reports results of crash risk regressions conditioned on information asymmetry. In Columns 1 and 3, the dummy variable DES equals one for firms with an above sample median of effective Bid-Ask spread, and zero otherwise. In Columns 2 and 4, the dummy variable DQS equals one for firms with an above sample median of percent quoted spread, and zero otherwise. DBN indicator is a dummy variable that equals one for the years in which DBN law is effective in a firm's state of headquarters, and zero otherwise. Other variables are as defined in Appendix 1. The models are estimated with firm and year fixed effects, but their estimates are suppressed for brevity. *t*-statistics based on heteroskedasticity-robust standard errors clustered by headquarters state are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

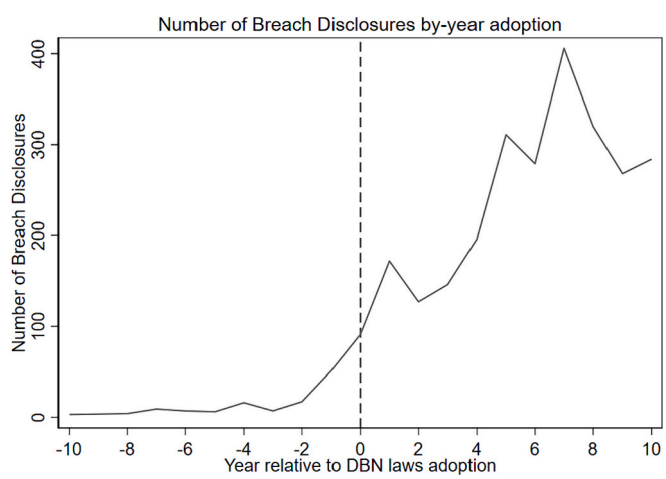

Fig. 2. DBN Laws and Number of Data Breach Disclosures by Year of Adoption.
This figure plots the number of breach disclosures over the 21-year window period centered on the year of DBN law adoption ($t = 0$). We collect data on data breaches from Privacy Rights Clearinghouse (PRC). Following Kamiya et al. (2021), we exclude incidents on governments, educational institutions, and non-profit organizations and focus only on breaches due to hacking or malware-electronic made by an outside party.

the interaction terms between each governance measure and DBN laws are positive and significant, suggesting that the positive impact of DBN laws on crash risk is more pronounced for firms with weaker corporate governance. These findings are consistent with our prediction.

5.2. Financial constraints and crash risk

In this section, we are interested in discerning whether firms' financial condition measured by their levels of financial constraints can affect the relation between DB laws adoption and crash risk. Despite the debate over the effectiveness of DBN laws on mitigating the risk of identity theft, firms appear to increase data security investments in the wake of disclosure laws (Law & Technology & Public Policy Clinic., 2007). However, Boasiako & Keefe (2021) report that financially constrained firms reduce investment while increasing cash reserves in response to the adoption of DBN laws. Because financially constrained firms face a higher cost of external financing (Korajczyk & Levy, 2003), these firms may lack adequate resources to support investments including those in data security. By contrast, financially unconstrained firms should be able to fund investment in data security as needed. To the extent that inadequate investment in data security exposes firms to more data breaches, we expect the positive effect of DBN laws adoption on crash risk to be stronger for financially constrained firms.

We use two different measures of financial constraints including firm size and SA_INDEX (Hadlock & Pierce, 2010). The SA_INDEX is calculated as $-0.737*Size + 0.043*Size^2 - 0.040*Age$, where Size equals the logarithm of inflation-adjusted book value of assets, and Age is the number of years the firm is included in Compustat. In calculating the index, we follow Hadlock and Pierce and cap the book value of assets at

Table 11
Data breach disclosures and abnormal stock returns.

Panel A: ARs and CARs around Data Breach Disclosures.

Day Relative to Event	AR	Cross-sectional t-stat for AR	CAR	Cross-sectional t-stat for CAR
−3	0.122	0.72	0.122	0.72
−2	−0.449*	−1.77	−0.327	−1.29
−1	−0.013	−0.08	−0.340	−1.17
0	−0.417*	−1.77	−0.757**	−2.16
1	−0.193	−1.17	−0.950**	−2.53
2	0.016	0.08	−0.932**	−2.12
3	0.121	0.63	−0.813*	−1.78

Panel B: CARs (in Percent) in Different Event Windows.

Model	CAR [−1,1]	CAR [−2,2]	CAR [−3,3]
	(1)	(2)	(3)
MM	−0.62**	−1.06**	−0.81*
	(2.00)	(2.22)	(1.78)

This table presents the average abnormal stock returns (ARs) and cumulative abnormal stock returns (CARs) for 65 data breach disclosures during the period 2005–2014. Estimations are based on the market model. Panel A shows ARs and CARs around data breach disclosure days. Panel B reports CARs (in percent) over different event windows (1−,1), (−2, 2), and (−3, 3), where days 0 is the data breach disclosure day. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

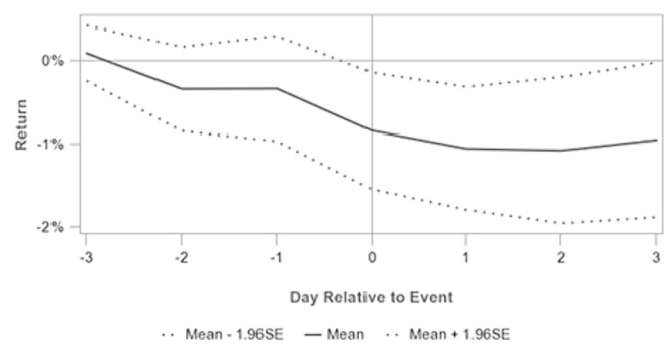

Fig. 3. Cumulative Abnormal Returns (in Percent) to Data Breach Announcement.
This figure plots the CARs together with 95% confidence intervals from 3 days before the data breach announcement to 3 days afterward. Estimations are based on the market model for the pre-event period of [−252,30].

$4.5 billion and firm age at 37 years. We categorize firms as financially constrained if their SIZE (SA_INDEX) is below (above) its sample median and run analysis for these subgroups separately. The results reported in Panel A of Table 9 indicate that the coefficients of DBN laws variable are positive but only significant for financially constrained firms. We further run crash risk regressions with interactions between the DBN laws indicator and financial constraint measures. Consistent with our expectation, the estimation results reported in Panel B of Table 9 indicate that the effect of the adoption of DBN laws on crash risk is more pronounced for financially constrained firms.

5.3. Information asymmetry and crash risk

Kothari et al. (2009) argue that information asymmetry facilitates bad news hoarding by managers. To examine whether the relation between DBN laws and crash risk varies with information asymmetry, we run triple difference regressions using effective bid-ask spread and percent quoted bid-ask spread as proxies for information asymmetry between informed and uninformed investors (Goyenko, Holden, & Trzcinka, 2009). A higher value of effective spread (percent quoted spread) implies a higher degree of information asymmetry. We create the dummy variable DES (DQS) that equals 1 for firms with effective spread (percent quoted spread) value above its sample median, and 0 otherwise. The results of the crash risk regressions reported in Table 10 indicate that the coefficients of the interactions between DBN laws and each measure of information asymmetry are positive and statistically significant, suggesting that the effect of DBN laws on crash risk is stronger for firms with a higher degree of information asymmetry.

Hutton et al. (2009) document that firms' discretionary earnings management is positively related to stock price crash risk. Since firms have to disclose bad news related to data breaches following the passage of DBN laws, they may engage in earnings management and other behaviors to offset the possible adverse effect of data breach disclosures on firm performance, which heightens crash risk. To explore this possible alternative explanation for the relation between DBN laws adoption and crash risk, we examine the effects of the DBN laws adoption on discretionary earnings management but do not find significant results. Thus, there is little evidence in support of the argument that firms increase earnings management following the DBN laws adoption, leading to higher crash risk.

5.4. Additional analysis

From a financial market perspective, the disclosure of bad news related to data breaches may trigger negative market reactions (Gatzlaff & McCullough, 2010), increasing crash risk. It is noteworthy that the adoption of DBN laws is likely to lead to an increase in disclosures of not only new data breaches as they occur but also the old ones withheld by managers, which intensifies crash risk. To explore this possibility, we collect data on data breaches from Privacy Rights Clearinghouse (PRC) and provide a graphical presentation of the number of data breaches disclosures in the 21-year window centered on DBN laws adoption years ($t = 0$) in Fig. 2. The graph in Fig. 2 shows that the number of data breach disclosures increases sharply immediately following the passage of DBN laws then declines gradually but remains at a significantly higher level than that before the adoption of DBN laws.

The adoption of DBN laws may affect not only crash risk but also other types of risk, such as downside risk (Ang et al., 2006), tail risk (Kelly & Jiang, 2014), bear beta (Lu & Murray, 2019), left-tail momentum (Atilgan et al., 2020), crash sensitivity (Chabi-Yo et al., 2018; Chabi-Yo et al., 2022). Therefore, in an additional analysis, we assess the stock price reaction to data breach disclosures, particularly following the adoption of DBN laws.[14] We run an event study of data breach disclosures (Acquisti, Friedman, & Telang, 2006; Foerderer & Schuetz, 2022) and calculate the daily abnormal stock returns (ARs) and cumulative abnormal stock returns (CAR) over an event window. Specifically, we use stock returns data during the period (−252, −30) preceding the data breach disclosure day to estimate the parameters of the market model and calculate the seven-day cumulative abnormal stock return (CAR) centered on a firm's data breach disclosure day. The event window begins three days before the event to account for a possible leakage of information. A firm may experience more than one data breach incident during the sample period; therefore, we keep only the first data breach of each firm during the sample period. Moreover, we drop events with undisclosed data breach scope and those coinciding with other corporate events (e.g., earnings announcement, M&As, capital increases, investment or disinvestments announcements, public offerings, rating actions, and restructuring announcements) during the event window. Our final sample includes 65 events over the period 2005–2014, which is in line with the number of data breach announcements reported in prior studies (e.g., Rosati et al., 2017).

Table 11 reports the ARs and CARs for the event window (−3,3). We find that ARs on days −2 and 0, where day 0 is the data breach disclosure day, are negative (−0.449%, −0.417%, respectively) and significant. In addition, CARs from day 0 and after also are negative and significant. Panel B of Table 11 reports CARs for different event windows (−1,1), (−2,2), and (−3, 3). On average, data breach disclosure CARs range from −1.06% to −0.62% and are statistically significant, indicating a large negative impact of data breach announcement on stock prices. We plot the CARs together with 95% confidence intervals in Fig. 3.[15] Although we cannot rule out other types of risk completely, our event study results suggest that data breach disclosures are more likely to be associated with crash risk than other types of risk.

6. Conclusion

Data breaches are a rising concern that has substantial consequences for businesses, consumers, and investors. To avoid harmful effects of data breach disclosures on their reputation, job security, and compensation, self-interested managers may withhold bad news about data breaches or time their disclosures opportunistically, thus, the effect of data breach disclosures on crash risk is unclear *ex ante*. Exploiting the staggered adoption of the DBN laws by U.S. states, which obligate firms to notify affected customers and involved parties of the cyber incidents when they occur, as a quasi-natural experiment, we document that the adoption of these laws is positively related to crash risk. Our results are robust to various model specifications, sample construction, and alternative measures of crash risk. Furthermore, our evidence indicates that the relation between DBN laws and crash risk varies with corporate governance, financial constraints, and information asymmetry. Overall, our findings suggest that data breach disclosures raise investors' concern about their consequences and vulnerability of data security of the breached firms, leading to higher crash risk.

Declarations of competing interest

None.

[14] We thank an anonymous referee for suggestion to run this analysis.

[15] In an unreported analysis, we run a cross-sectional regression of the data breach disclosure CARs on the DBN law indicator variable and other controls. Although the regression sample is smaller due to missing observations of some control variables, we find that the negative effect of data breach disclosures on abnormal stock returns is more pronounced following the DBN laws adoption that curbs managerial discretion.

Data availability

The authors do not have permission to share data.

Appendix

Appendix 1

Variables definition.

Variable	Description
Data breach notification laws	
DBN_t	Indicator variable set to one if a firm's state of headquarters has adopted a data breach notification law, and zero otherwise.
Crash risk variables	
$NCSKEW_{t+1}$	Negative skewness of daily firm-specific stock returns over the fiscal year t + 1. See Eq. (2).
$DUVOL_{t+1}$	The log of the ratio of the standard deviations of down-day to up-day daily returns over the fiscal year t + 1. See Eq. (3).
$CRASH_COUNT_{t+1}$	The difference in the number of times firm-specific daily returns is above, to those that are below, 3.09 standard deviations of the mean, where 3.09 is set to generate frequencies of 0.01% in the log-normal distribution.
$CRASH_{t+1}$	An indicator variable that equals one for a firm-year that experiences one or more crash (firm-specific daily returns falling 3.09 or more standard deviations below its mean daily return) during the fiscal year period, and zero otherwise.
$EXTRASIGMA_{t+1}$	The extreme sigma, measured as the negative of the worst deviation of firm-specific daily returns from the average firm-specific daily return divided by the standard deviation of firm-specific daily returns in a fiscal year.
Baseline variables	
$DTURN_t$	The average monthly share turnover over the fiscal year *t* minus the average monthly share turnover over the previous fiscal year t-1, where monthly share turnover is calculated as the monthly trading volume divided by the total number of shares outstanding during the month.
RET_t	The mean of firm-specific daily returns over the fiscal year, times 100.
$SIGMA_t$	The standard deviation of firm-specific daily returns over the fiscal year.
$SIZE_t$	The natural logarithm of market capitalization (csho×prcc_f) at the end of the fiscal year.
MB_t	The market value of equity (csho×prcc_f) divided by the book value of equity (ceq).
LEV_t	The book value of all liabilities (dltt + dlc) scaled by total assets (at) at the end of the fiscal year.
ROA_t	Income before extraordinary items (ib) divided by lagged total assets (at_lag).
$OPAQUE_t$	The absolute value of discretionary accruals averaged over the prior three years, where discretionary accruals are estimated by the modified Jones model (Dechow et al., 1995).
State-level variables	
PCI_t	Annual per capital income ($ mil.)
GSP_t	Annual gross state product ($ mil.)
$UEMP_t$	Annual state level unemployment rate (%)
ΔGSP_t	3-year average growth rate in gross state production (%)
Other variables	
$INST_OWN_t$	A dummy variable that equals one for firms with below sample median institutional ownership in each fiscal year, and zero otherwise.
ANA	A dummy variable that equals one for firms with below sample median analyst coverage in each fiscal year, and zero otherwise.
AGE	Number of years the firm is included in Compustat.
SA_INDEX_t	is constructed following Hadlock and Pierce (2010) as $-0.737*Size + 0.043*Size^2 - 0.040*Age$, where Size equals the log of inflation adjusted Compustat item at (in 2004 dollars), and Age is the number of years the firm is listed with a non-missing stock price on Compustat. In calculating the index, we follow Hadlock and Pierce and cap Size at $4.5 billion and Age at 37 years. A dummy variable that equals one for firms with above sample median SA index in each year, and zero otherwise.
GEO_DISP	Geographic dispersion is measured as the number of U.S. states mentioned in the annual report filed on Form 10-K with SEC.
ES	$ES_k = 2*\lvert \ln(P_k) - \ln(M_k) \rvert$, where P_k is the price of the kth trade and M_k is the midpoint of the consolidated best bid and offer (BBO) prevailing at the time of the kth trade. The effective spread is then averaged over the fiscal year (Goyenko et al., 2009).
QS	$QS = \frac{As - Bs}{Ms}$, where A_s is the National Best Ask and B_s is the National Best Bid, and M_s is the bid-ask midpoint and calculated by taking the average of A_s and B_s. The percent quoted spread is then averaged over the fiscal year.
$Before^{2+}$	An indicator variable that takes one for observations with two years or more prior to the adoption of DBN laws, and zero otherwise.
$Before^{1}$	An indicator variable that takes one for observations with one year prior to the adoption of DBN laws, and zero otherwise.
$After^{1}$	An indicator variable that takes one for observations with one-year after the adoption of DBN laws, and zero otherwise.
$After^{2+}$	An indicator variable that takes one for observations with two years or after the adoption of DBN laws, and zero otherwise.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.irfa.2024.103164.

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